1. a. $f(x) = \frac{a}{\sigma}\left(\frac{x}{\sigma}\right)^{a-1} \cdot \exp\left(-\left(\frac{x}{\sigma}\right)^{a}\right) \qquad x>0$

$$F(x) = \int_0^x \frac{a}{\sigma}\left(\frac{t}{\sigma}\right)^{a-1} \cdot e^{-\left(\frac{t}{\sigma}\right)^a} dt \qquad u = \left(\frac{t}{\sigma}\right)^a = \frac{t^a}{\sigma^a}$$

$$du = \frac{a t^{a-1}}{\sigma^a} dt$$

$$F(x) = \int_0^x \frac{a}{\sigma} \cdot \frac{t^{a-1}}{\sigma^{a-1}} \cdot e^{-\left(\frac{t}{\sigma}\right)^a} dt$$

$$F(x) = \int_0^x \frac{a t^{a-1}}{\sigma^a} \cdot e^{-\left(\frac{t}{\sigma}\right)^a} dt$$

$$F(x) = \int_0^{\frac{x^a}{\sigma^a}} e^{-u}\, du$$

$$= \left[-e^{-u}\right]_0^{\frac{x^a}{\sigma^a}}$$

$$= \left[-e^{-\frac{x^a}{\sigma^a}} + 1\right]$$

$$= 1 - \exp\left[-\left(\frac{x}{\sigma}\right)^a\right]$$

b. Survival Function

$$S(x) = 1 - F(x)$$
$$= 1 \cancel{+1} + \exp\left[-\left(\frac{x}{\sigma}\right)^a\right])$$
$$= \exp\left[-\left(\frac{x}{\sigma}\right)^a\right]$$

c.

$$h_0(t) = \frac{\frac{a}{\sigma}\left(\frac{x}{\sigma}\right)^{a-1} \cdot \cancel{\exp\left(-\left(\frac{x}{\sigma}\right)^a\right)}}{\cancel{\exp\left(-\left(\frac{x}{\sigma}\right)^a\right)}} = \frac{a}{\sigma} \cdot \frac{x^{a-1}}{\sigma^{a-1}} = \frac{a x^{a-1}}{\sigma^a}$$

$$h(t) = h_0(t) \cdot \exp(x^T\beta) = \frac{a x^{a-1}}{\sigma^a} \cdot \exp(x^T\beta)$$

3. b.